

METHODIST UNIVERSITY GHANA



FACULTY OF SCIENCE

DEPARTMENT OF NURSING AND MIDWIFERY

END OF SECOND SEMESTER EXAMINATION: 2024/2025

COURSE CODE/TITLE: MNUR 102, ANATOMY AND PHYSIOLOGY II

CAMPUS: DANSOMAN AND TEMA (REGULAR AND WEEKEND)

3 CREDITS: TIME ALLOWED: 2HRS 15MINS

INSTRUCTIONS:

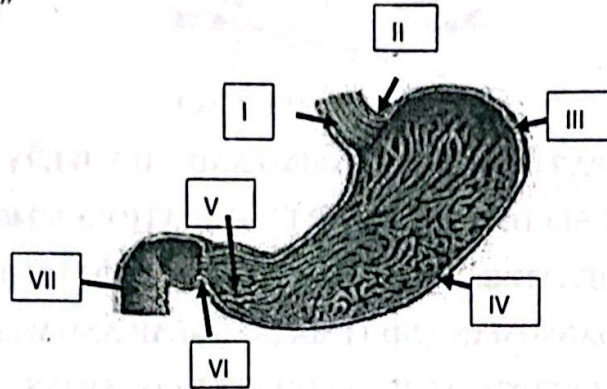
1. Do not open this booklet until you are told to do so
2. For the section A, answer all questions in this booklet
3. Choose the appropriate options A-C and indicate in the spaces provided on the right-hand side of each question
4. For section B, provide your answers in the answer booklet given you
5. All forms of examination malpractices will be duly punished
6. Ensure you have been well searched, leave all items including phones, smart watches and any unprescribed electronic gadget out of the hall
7. In the situation where you think there is a typing error or there is no answer to a given question, put a question mark against the question, it will be addressed when marking of scripts

Examiner: Stephen Asante Obeng

Page 1 of 11

SECTION A: FOR QUESTIONS 1-60, CHOOSE FROM THE APPROPRIATE OPTIONS A-C THE ONE THAT IS THE MOST APPROPRIATE ANSWER
TIME ALLOWED: 1HR 15MINS

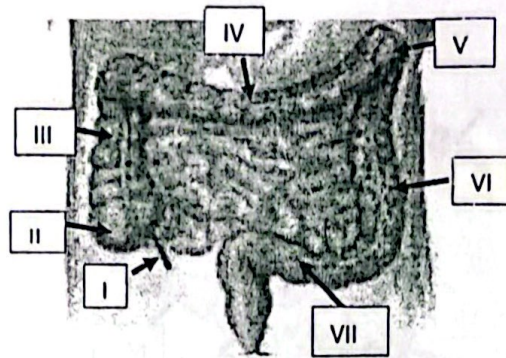
"Mr. U, a 50yr old blacksmith, is being assessed for the effect of alcohol ingestion on his body. During endoscopy, photographs of body parts involved were taken to help students appreciate anatomical structure"



Study the image carefully and answer questions 1-5.

1. The process of viewing the internal structure of the organ in this scenario is called?
A. Duodenoscopy
B. Gastrosocopy
C. Oesophagoscopy
Ans.....
2. The part labelled "II" represents which structure?
A. Lower oesophageal Sphincter
B. Pyloric sphincter
C. Upper oesophageal sphincter
Ans.....
3. In order to perform the test described in question (1), it is important to keep the organ empty. The best way to ensure this is to?
A. Ensure patient fasts at least 6 hours before the procedure
B. Ensure patient takes only liquid foods few hours before the procedure
C. Ensure patient takes oral medication with little amount of water the morning of the procedure
Ans.....
4. When performing the test in question (1), particular attention is given to the region where food is mainly stored (body). Which of the labels corresponds to this part?
A. III
B. IV
C. V
Ans.....
5. After the test in (1), the patient complained of chest tightness, heart burns and bitter taste in the mouth, possible signs of regurgitation. What intervention by the nurse will minimise these symptoms?
A. Keep patient in semi-fowlers position at least 1 hour after the test
B. Place patient in the supine position to promote comfort
C. Place patient in the Trendelenburg position to improve venous return
Ans.....

Madam Y, a 45 yr old woman, has been scheduled for exploratory laparotomy. As part of preparation for surgery, an ultrasound was taken to examine structures of the abdomen. The diagram below is a cartoon version of the scan saved for anatomical studies.



Study the image above carefully and answer the questions 6-10.

6. Which organ in the abdomen has been labelled in the diagram?

A. Colon
B. Intestines
C. Ileum

Ans.....

7. After reviewing the scan, an ulcer (wound) was found in a certain part of the organ under study. To investigate the ulcer further, a sigmoidoscopy was requested. The nurse explains to the patient that the procedure involves insertion of an endoscope for the purpose of viewing which part of the organ?

A. V
B. VI
C. VII

Ans.....

8. After the test in question (7), an ulcer of size 1cm x 0.5cm was confirmed. What information given by the patient confirms this finding?

A. Passage of bloody stool for the past 2 months
B. Passage of pale stool for the past 2 months
C. Passage of brown stools for the past 2 months

Ans.....

9. The surgeon decided to remove the ulcer (colectomy) and create an opening into the part labelled VI (colostomy) to allow for passage of stool. The type of stool in VI often has this characteristic?

A. Paste
B. Watery
C. Well-formed (solid)

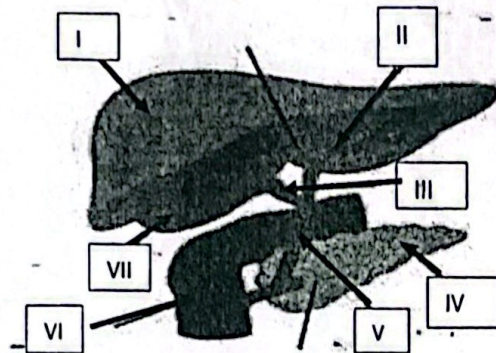
Ans.....

10. Before the surgical procedure in (9), the nurse administers an enema containing an antibiotic preparation? The purpose of bowel preparation is to reduce the number ofin the large bowel!

A. Escherichia coli
B. Lactobacillus acidophilus
C. Staphylococcus aureus

Ans.....

Miss T reported to the hospital with history of pale stools and abdominal pains at 2-4 hours after eating fat foods. A Magnetic Resonance Imaging (MRI) of relevant part has been shown in the figure below.



Study the image carefully and answer the questions 11-15

11. Which organ should be assessed further if gall stones are suspected?

- A. I
- B. IV
- C. VII

Ans.....

12. The pale stool indicate a problem with the flow of?

- A. bile
- B. intestinal juice
- C. pancreatic juice

Ans.....

13. In order assess the state of structure I, the nurse palpates which part of the abdomen?

- A. Epigastric
- B. Right upper quadrant
- C. Left upper quadrant

Ans.....

14. What is the appropriate name given to the system in the figure above?

- A. Biliary
- B. Lower GIT
- C. Upper GIT

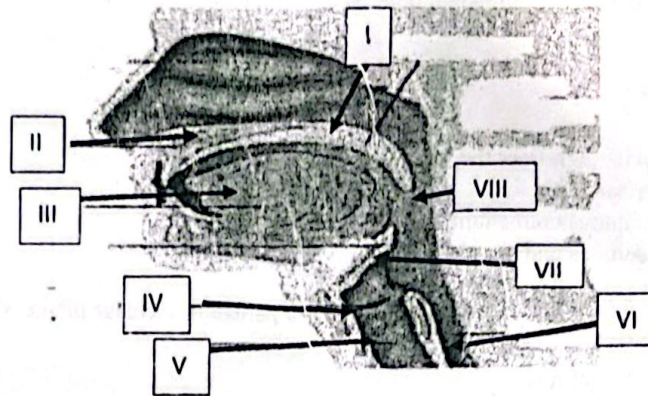
Ans.....

15. The nurse tries to explain to the patient why pains are severe after eating fatty foods. Which of the following is accurate?

- A. Fats stimulate the release of bile from the gall bladder which worsens the condition
- B. Fats irritate the mucosa of the duodenum resulting in painful signs of inflammation
- C. Fats are associated with cancers of the duodenum, hence the pain

Ans.....

A patient reported to the hospital with a complain of coughing every time he attempts to swallow food. The image below is a cartoon version of a Computerised Tomography (CT) scan of the relevant structures.



Use the images above to answer questions 16-20

16. The appropriate name given to the region captured by the CT scan is the.....

- A. lower GIT
- B. oral cavity and pharynx
- C. upper GIT

Ans.....

17. After swallowing, food changes name to.....

- A. bolus
- B. chyme
- C. nutrients

Ans.....

18. What is the normal path of food during swallowing?

- A. VIII to VI
- B. VII to IV, to V
- C. VI, to IV, to V

Ans.....

19. The coughing is most likely to be caused by failure of structureto cover the airway during swallowing.

- A. I
- B. V
- C. VII

Ans.....

20. What education would the nurse give to the patient to minimise coughing during swallowing?

- A. Avoid drinking whilst eating
- B. Avoid writing whilst eating
- C. Avoid talking whilst eating

Ans.....

21. There are three phases of swallowing, oral, pharyngeal and
A. Duodenal
B. Gastric
C. Oesophageal
Ans.....
22. Which type of epithelium lines the lower part of the oesophagus?
A. Simple columnar epithelium
B. Stratified squamous epithelium
C. Stratified squamous and simple squamous epithelia
Ans.....
23. To maintain a healthy tooth, the nurse encourages a patient to increase intake of
A. Calcium and Vitamin C
B. Flouride and Vitamin K
C. Proteins and Phosphates
Ans.....
24. Which of the following is **UNTRUE** about the teeth
A. They erupt from the mandibles and maxillary sockets
B. The outermost part is made up of dead cells
C. The nerves and blood vessels are found in the dentine layer
Ans.....
25. Functions of the normal flora in the colon include synthesis of which food nutrient?
A. Vitamin B12
B. Vitamin C
C. Vitamin K
Ans.....
26. Structural adaptations of the ileum include which of the following?
A. Less Convolutions
B. Small surface area
C. Villi and microvilli
Ans.....
27. Adaptations of the GIT to fight ingested bacteria include which of the following?
A. Highly acidic gastric juice
B. Presence of lymphoid aggregates in the mouth and intestines
C. Both A and B
Ans.....
28. A patient has been passing dark stool for the past 3 days. The stool is likely to contain
A. blood
B. too much cholesterol
C. too much fibre
Ans.....
29. In the intestines, water soluble nutrients such as glucose are absorbed into the
A. lacteals
B. liver
C. mesenteric capillaries
Ans.....
30. After gastric surgery, a patient is given intravenousdue to possible damage to theduring surgery
A. Vitamin B12, parietal cells
B. Vitamin C, chief cells
C. Vitamin K, goblet cells
Ans.....